CONFIDENTIAL: contains the reference answer and rubric. Never place this document where a model can read it.

dlbcl_0002 · v2 case pack

Nodal and splenic DLBCL after prior anthracycline for breast cancer, reduced LVEF: tumor-board plan

Difficulty: advanced · Chart exported 2026-09-17 · Review status: **PENDING_ONCOLOGIST** · Harbor tasks: dlbcl_0002_splenic_stage_iii_cardiac_v2_narrative, dlbcl_0002_splenic_stage_iii_cardiac_v2_fhir

Item	Summary
Patient	59-year-old woman, ECOG 1. Axillary, mediastinal, mesenteric and para-aortic nodes with splenic nodules; fevers and night sweats.
Core teaching point	Splenic involvement with nodes on both sides of the diaphragm is stage III, and the spleen is not an extranodal IPI site (IPI 2).
Real-world twist	Breast cancer in 2016 treated with doxorubicin (240 mg/m² cumulative) and left chest radiotherapy. LVEF has fallen from 62% to 45%, GLS -14%, with no heart-failure symptoms.
Key decisions	• Cancer therapy-related cardiac dysfunction: do not give standard full-dose doxorubicin (dangerous to ignore) • Urgent cardio-oncology review; ACE inhibitor/ARB plus beta-blocker • Curative anthracycline-free regimen: R-CEOP preferred, R-GCVP as alternative; R-CHOP with liposomal doxorubicin or dexrazoxane only with cardio-oncology agreement • Serial LVEF and troponin/NT-proBNP; end-of-treatment PET-CT
For the oncologist	• Which regimen does the oncologist prefer: R-CEOP, R-GCVP, or cardioprotected R-CHOP? • Should cardiac MRI or an ischemic work-up be required? • Anastrozole: continuing it and holding it are both accepted.

Synthetic training case for AI evaluation. Not clinical decision support. No real patient data.

A. Task instruction (model-visible)

````
You are the consulting hematologist-oncologist for this patient.

This is a synthetic training case for AI evaluation. It is not clinical decision support.

The chart export is in `/app/case/chart/`: a patient summary followed by dated notes, reports and
laboratory results, numbered in chronological order.

The patient will be presented at lymphoma tumor board tomorrow. Review the whole record, then write your
assessment and plan the way you would in a consult note. Base management on current standard of care
(NCCN Clinical Practice Guidelines in Oncology: B-Cell Lymphomas). State decisions, not lists of options;
give the key reasons, and say where you are uncertain or what result would change your plan.

Write your answer to:

```text
/app/final_answer.json
```

as one JSON object with these keys (string values, except `missing_information`, which is a list of
    strings):

- `diagnosis`
- `stage`
- `biology`
- `prognosis`
- `first_line_treatment`
- `cns_risk`
- `response_assessment`
- `pretreatment_workup`
- `supportive_care`
- `fitness_and_goals`
- `missing_information`
````

B. Patient chart (model-visible, narrative rendering)

The FHIR rendering carries the same facts, generated from the same source file (cases_src/dlbcl_0002/case.toml). Files appear in chronological order, as the model sees them.

chart/00_patient_summary.txt

```
PATIENT SUMMARY (EHR export 2026-09-17)

Name: Grace Sample (synthetic)    MRN: OW-0002
DOB: 1967-03-08 (59 y)    Sex: female

PROBLEM LIST
- Malignant neoplasm of left breast, ER-positive, HER2-negative (since 2016). Stage IIA (pT2 pN0).
    Lumpectomy and sentinel node biopsy, adjuvant chemotherapy (doxorubicin/cyclophosphamide then
    paclitaxel), left breast radiotherapy; on anastrozole since November 2016. No evidence of recurrence at
    last surveillance visit (March 2026).
- Essential hypertension (since 2018). On amlodipine.
- Osteopenia (since 2019). On aromatase inhibitor. DEXA 2025: lumbar spine T-score -1.8.

ACTIVE MEDICATIONS
- Anastrozole 1 mg tablet, 1 tablet by mouth daily (since November 2016)
- Amlodipine 5 mg tablet, 1 tablet by mouth daily
- Calcium carbonate 600 mg / vitamin D3 800 IU tablet, 1 tablet by mouth twice daily
- Acetaminophen 500 mg tablet, 1-2 tablets by mouth every 6 hours as needed for fever (patient-reported,
    started August 2026)

ALLERGIES
- Penicillin: Hives (childhood)

This is a synthetic training case for AI evaluation. It is not clinical decision support.
```

chart/01_2026-03-10_progress_note.txt

```
BREAST ONCOLOGY SURVIVORSHIP VISIT
Date: 2026-03-10    Author: Synthetic Breast Medical Oncologist

Ms. Sample returns for annual survivorship follow-up, now 9.5 years from diagnosis of left breast
cancer.

Oncology history:
- 2016-01: screen-detected left breast mass, 10 o'clock. Core biopsy: invasive ductal carcinoma,
  grade 2, ER 95%, PR 60%, HER2 1+ (negative).
- 2016-02: left lumpectomy and sentinel lymph node biopsy (2 left axillary sentinel nodes removed).
  pT2 (2.4 cm) pN0 (0/2), margins negative. Stage IIA.
- 2016-03 to 2016-07: adjuvant dose-dense doxorubicin 60 mg/m2 plus cyclophosphamide 600 mg/m2 every
```

  2 weeks x 4 cycles (cumulative doxorubicin 240 mg/m2), then weekly paclitaxel 80 mg/m2 x 12.
  Pre-chemotherapy echocardiogram 2016-02-15: LVEF 62%. No echocardiogram since.
- 2016-08 to 2016-10: whole-breast radiotherapy to the left breast, 50 Gy in 25 fractions plus a
  10 Gy tumor-bed boost.
- 2016-11 to present: anastrozole 1 mg daily. Extended therapy to 10 years planned, to finish
  November 2026.

Interval history: tolerating anastrozole with mild arthralgias. No breast changes, bone pain, cough,
or headache. Energy good; attends exercise classes three times a week.

Exam: BP 138/84, HR 76. Weight 72.8 kg. Healed left lumpectomy scar with mild radiation
skin changes. No breast masses. No axillary, supraclavicular or cervical lymphadenopathy.
Heart regular, no murmur. Lungs clear. No edema.

Diagnostic bilateral mammogram 2026-03-02: post-lumpectomy changes on the left, stable. BI-RADS 2.

Plan: continue anastrozole to complete 10 years in November 2026. Annual mammogram. DEXA due 2027.
Return in 12 months.

chart/02_2026-03-10_labs.txt

LABORATORY RESULTS: Breast oncology survivorship chemistry
Collected: 2026-03-10   Synthetic Clinical Laboratory

Test	Result	Flag	Units	Reference
Creatinine	0.8		mg/dL	0.5-1.1
eGFR (CKD-EPI 2021)	85		mL/min/1.73m2	>=60
Alkaline phosphatase	88		U/L	40-129
LDH	182		U/L	135-225

chart/03_2026-08-12_referral.txt

PRIMARY CARE REFERRAL: LEFT AXILLARY MASS
Date: 2026-08-12   Author: Synthetic Primary Care Physician

Reason for referral: new left axillary mass in a breast cancer survivor; recurrence vs other.

Ms. Sample is a 59-year-old school librarian with a history of left breast cancer (2016), now on
anastrozole, and hypertension. Noticed a firm, non-tender lump in the left armpit in mid-July.
Over the past 4 weeks she has had evening fevers, which she has been recording at home, soaking
night sweats two to three times a week, and fatigue. Weight down about 2 kg.

Left axillary ultrasound 2026-08-05: several enlarged left axillary lymph nodes, largest 3.4 cm, with
loss of the fatty hilum.

CT chest/abdomen/pelvis with IV contrast 2026-08-07 (outside facility): enlarged left axillary and
mediastinal lymph nodes; mesenteric and para-aortic lymphadenopathy, largest mesenteric conglomerate
7.4 cm; mildly enlarged spleen (12.6 cm) with several hypodense lesions up to 2 cm. Liver, adrenal
glands and kidneys normal. Impression: lymphoma favored; metastatic breast carcinoma not excluded.

Referred to breast surgical oncology for excisional biopsy and to hematology-oncology.

chart/04_2026-08-28_pathology.txt

SURGICAL PATHOLOGY: EXCISIONAL BIOPSY, LEFT AXILLARY LYMPH NODE
Date: 2026-08-28   Author: Synthetic Hematopathologist
Status: amended 2026-09-08

SPECIMEN: Left axillary lymph node, excisional biopsy, collected 2026-08-25. Received fresh; portions
submitted for flow cytometry and cytogenetics.
CLINICAL HISTORY: Left breast carcinoma 2016 (ER-positive, HER2-negative). New left axillary
lymphadenopathy, fevers.

GROSS: Lymph node, 3.6 x 2.8 x 2.0 cm, tan-white fleshy cut surface with focal necrosis.

FINAL DIAGNOSIS: DIFFUSE LARGE B-CELL LYMPHOMA, NOT OTHERWISE SPECIFIED.

MICROSCOPIC: Effacement of nodal architecture by diffuse sheets of large atypical lymphoid cells with
vesicular chromatin and one or more nucleoli; numerous mitoses and focal necrosis. No cohesive
epithelial nests, glands or single-file infiltrating epithelial cells.

IMMUNOHISTOCHEMISTRY (adequate controls):

CD20	positive (diffuse, strong)
PAX5	positive
CD3	negative in tumor cells (scattered background T cells)
CD5	negative
CD10	negative
BCL6	positive
MUM1/IRF4	positive (approximately 80% of tumor cells)
BCL2	positive in approximately 70% of tumor cells
MYC	positive in approximately 45% of tumor cells
Cyclin D1	negative
CD30	negative
Ki-67	approximately 80%

```
  EBER ISH          negative
  Cytokeratin AE1/AE3   negative in lesional cells
  GATA3             negative in lesional cells

FLOW CYTOMETRY: CD19+, CD20+, CD10−, kappa light-chain restricted large B-cell population (62% of
events).

COMMENT: Non-germinal-center B-cell-like immunophenotype by the Hans algorithm (CD10 negative, BCL6
positive, MUM1 positive). Cytokeratin AE1/AE3 and GATA3 are negative in the lesional cells and there is
no morphologic evidence of metastatic carcinoma. Interphase FISH has been requested on block A2;
results will be reported in an addendum.

ADDENDUM (2026-09-08): FISH, interphase, break-apart probes, 200 nuclei scored.
  MYC (8q24):   no rearrangement detected
  BCL2 (18q21): no rearrangement detected
  BCL6 (3q27):  no rearrangement detected
The final diagnosis is unchanged.
```

chart/05_2026-09-01_consult.txt

```
HEMATOLOGY-ONCOLOGY NEW PATIENT CONSULTATION
Date: 2026-09-01    Author: Synthetic Hematologist-Oncologist

HPI: 59-year-old woman referred after excisional biopsy of a left axillary node (2026-08-25) showed
diffuse large B-cell lymphoma (FISH pending). She first noticed the left axillary lump in mid-July.
For about 5 weeks she has had evening fevers several times a week; her home log, brought today,
documents temperatures of 37.9 to 38.4 C, maximum 38.4 C on 2026-08-27, without infectious symptoms.
Drenching night sweats requiring a change of nightclothes two to three times a week. Fatigue: still
working her usual hours at the school library and independent in all activities, but has stopped her
exercise classes. Weight 70.5 kg today vs 72.8 kg in March 2026 (about 3% loss). No abdominal pain,
early satiety, bone pain, cough, headache, visual change, focal weakness or confusion.
Cardiovascular: no exertional dyspnea, orthopnea, paroxysmal nocturnal dyspnea, ankle swelling, chest
pain, palpitations or syncope; climbs two flights of stairs without stopping (NYHA class I).

ONCOLOGY HISTORY: Left breast invasive ductal carcinoma 2016, pT2 pN0, stage IIA, ER-positive,
PR-positive, HER2-negative. Lumpectomy and sentinel node biopsy (2016-02); dose-dense AC (doxorubicin
60 mg/m2 and cyclophosphamide 600 mg/m2, 4 cycles; cumulative doxorubicin 240 mg/m2) followed by weekly
paclitaxel x 12 (2016-03 to 2016-07); left whole-breast radiotherapy 50 Gy + 10 Gy boost (2016-08 to
2016-10); anastrozole since 2016-11, now in year 10, planned to stop November 2026. Last mammogram
2026-03 benign.

PMH: hypertension (2018) on amlodipine; osteopenia on calcium/vitamin D. No known coronary disease,
diabetes or arrhythmia. No other cardiac history.
Medications: see medication list.
Allergies: penicillin (hives).
Social: school librarian, married, lives with her husband 20 minutes from the cancer center. Never
smoker. Rare alcohol. Two adult children.
Family history: mother with heart failure in her 70s; father with colon cancer at 74. No known
BRCA testing; criteria were not met in 2016.

Exam: T 37.8 C, HR 84, BP 146/88, RR 14, SpO2 98% RA. Weight 70.5 kg, height 165 cm (BSA 1.79 m2).
Well-appearing. Healing left axillary incision; residual 2 cm firm left axillary node. No cervical,
supraclavicular or inguinal lymphadenopathy. Breasts: healed left lumpectomy scar, no masses
bilaterally. Heart: regular rhythm, no murmur or gallop; JVP not elevated. Lungs clear. Abdomen soft,
non-tender; spleen tip palpable 2 cm below the left costal margin; no hepatomegaly. No peripheral
edema. Neuro: non-focal.

ECOG performance status 1.

Plan today: CBC, CMP, LDH, uric acid, beta-2 microglobulin, viral serologies; FDG PET-CT;
transthoracic echocardiogram (prior doxorubicin and left breast radiotherapy); bone marrow biopsy;
await FISH. Return after results.
```

chart/06_2026-09-01_labs.txt

```
LABORATORY RESULTS: Complete blood count with differential
Collected: 2026-09-01    Synthetic Clinical Laboratory

Test                                  Result  Flag  Units           Reference
Hemoglobin                              11.4  L     g/dL            12.0-15.5
MCV                                       88        fL              80-100
WBC                                      6.9        x10^3/uL        4.0-11.0
Absolute neutrophils                     4.6        x10^3/uL        1.8-7.7
Absolute lymphocytes                     1.2        x10^3/uL        1.0-4.8
Platelets                                178        x10^3/uL        150-400
```

chart/07_2026-09-01_labs.txt

```
LABORATORY RESULTS: Comprehensive metabolic panel and tumor markers
Collected: 2026-09-01    Synthetic Clinical Laboratory

Test                                  Result  Flag  Units           Reference
Sodium                                   139        mmol/L          135-145
Potassium                                4.2        mmol/L          3.5-5.1
```

Chloride	103		mmol/L	98–107
CO2, total	25		mmol/L	22–29
BUN	14		mg/dL	7–20
Creatinine	0.8		mg/dL	0.5–1.1
eGFR (CKD-EPI 2021)	85		mL/min/1.73m2	>=60
Glucose	96		mg/dL	70–99
Calcium	9.3		mg/dL	8.6–10.3
Albumin	3.8		g/dL	3.5–5.0
Total protein	7.0		g/dL	6.0–8.3
AST	24		U/L	10–40
ALT	22		U/L	7–56
Alkaline phosphatase	92		U/L	40–129
Bilirubin, total	0.6		mg/dL	0.2–1.2
Uric acid	5.8		mg/dL	3.4–7.0
Phosphorus	3.6		mg/dL	2.5–4.5
LDH	510	H	U/L	135–225
Beta-2 microglobulin	2.9	H	mg/L	0.8–2.2

chart/08_2026-09-01_labs.txt

LABORATORY RESULTS: Infectious disease serologies
Collected: 2026-09-01 Synthetic Clinical Laboratory

Test	Result	Flag	Units	Reference
Hepatitis B surface antigen	Nonreactive			Nonreactive
Hepatitis B core antibody, total	Nonreactive			Nonreactive
Hepatitis B surface antibody	Nonreactive			Nonreactive
Hepatitis C antibody	Nonreactive			Nonreactive
HIV-1/2 Ag/Ab screen	Nonreactive			Nonreactive

chart/09_2026-09-04_pet_ct.txt

FDG PET-CT, SKULL BASE TO MID-THIGH
Date: 2026-09-04 Author: Synthetic Nuclear Medicine Physician

CLINICAL INDICATION: Diffuse large B-cell lymphoma on left axillary node excision (2026-08-25).
History of left breast cancer 2016. Initial staging.

TECHNIQUE: 9.8 mCi 18F-FDG IV; uptake time 60 minutes; serum glucose at injection 94 mg/dL.
Low-dose non-contrast CT for attenuation correction and localization. Mediastinal blood pool SUVmax
1.9; liver SUVmax 2.7. Comparison: outside CT 2026-08-07.

FINDINGS:
Head and neck: physiologic brain uptake (PET is insensitive for intracranial disease). No FDG-avid
cervical or supraclavicular lymph nodes.
Chest:
- Left axillary lymph nodes, largest 2.2 cm, SUVmax 17.8, adjacent to post-excision changes.
- Mediastinal lymph nodes (prevascular and left paratracheal), largest 2.9 cm, SUVmax 15.4.
- Left breast: post-lumpectomy changes without FDG-avid lesion. Right breast normal.
- Mild left anterior lung fibrotic change in the prior radiation field, not FDG-avid. No pulmonary
  nodules. No pleural or pericardial effusion. Physiologic myocardial uptake.
Abdomen and pelvis:
- Mesenteric nodal conglomerate, 7.5 x 5.1 cm, SUVmax 22.3 (largest nodal mass).
- Para-aortic lymph nodes, largest 3.0 cm, SUVmax 16.9.
- Spleen 12.8 cm with multiple FDG-avid nodules, largest 2.1 cm, SUVmax 16.2.
- Liver normal without focal uptake. Adrenal glands and kidneys normal without focal uptake.
- No FDG-avid pelvic or inguinal nodes. No ascites.
Musculoskeletal: no focal osseous or marrow FDG uptake.

IMPRESSION:
1. FDG-avid lymphadenopathy above the diaphragm (left axillary, mediastinal) and below the diaphragm
   (mesenteric, para-aortic), with multiple FDG-avid splenic nodules, consistent with lymphoma.
   Largest nodal mass 7.5 cm (mesenteric).
2. No FDG-avid lesion in the kidneys, adrenal glands, liver, lungs, bone, marrow or breasts.
3. Post-treatment changes of the left breast and left anterior lung without evidence of active disease.

chart/10_2026-09-09_echocardiogram.txt

TRANSTHORACIC ECHOCARDIOGRAM
Date: 2026-09-09 Author: Synthetic Cardiologist

Indication: baseline before planned lymphoma chemotherapy. Prior doxorubicin (2016) and left breast
radiotherapy (2016). Hypertension.
BP 148/90, HR 82, sinus rhythm.

Left ventricle: normal size (LVIDd 4.9 cm), normal wall thickness. Mild global hypokinesis without a
regional wall-motion abnormality. LVEF 45% (biplane Simpson's). Global longitudinal strain -14%
(laboratory normal more negative than -18%). Grade I diastolic dysfunction.
Right ventricle: normal size and function (TAPSE 2.1 cm).
Valves: mild mitral regurgitation; aortic valve trileaflet without stenosis.
No pericardial effusion. Estimated RV systolic pressure 28 mmHg.

Comparison: transthoracic echocardiogram 2016-02-15, LVEF 62% (biplane), normal wall motion; strain not
performed.

CONCLUSION:
1. Mildly reduced left ventricular systolic function, LVEF 45%, with mild global hypokinesis and
   reduced global longitudinal strain (−14%).
2. LVEF decreased from 62% on the 2016 study.
3. Grade I diastolic dysfunction. Mild mitral regurgitation.

chart/11_2026-09-10_bone_marrow.txt

BONE MARROW ASPIRATE AND BIOPSY, RIGHT POSTERIOR ILIAC CREST
Date: 2026-09-10   Author: Synthetic Hematopathologist

Collected 2026-09-08. Core length 2.0 cm, adequate. Normocellular marrow for age (about 40%) with trilineage hematopoiesis. No lymphoma identified on morphology or CD20/PAX5 immunostains. Flow cytometry of the aspirate: no monotypic B-cell population. Iron stores adequate.
DIAGNOSIS: No involvement by lymphoma.

chart/12_2026-09-17_progress_note.txt

HEMATOLOGY-ONCOLOGY FOLLOW-UP (RESULTS VISIT, PRE-TUMOR BOARD)
Date: 2026-09-17   Author: Synthetic Hematologist-Oncologist

Results reviewed with the patient and her husband: PET-CT, bone marrow, FISH addendum and echocardiogram. Evening fevers continue (38.2 C two nights ago) with night sweats; taking acetaminophen. Weight 70.1 kg. Still working; fatigue unchanged. No dyspnea, orthopnea, edema or chest pain. No neurological symptoms.

Exam: T 37.6 C, HR 86, BP 148/90. Residual left axillary node about 2 cm; spleen tip palpable. Lungs clear, no edema, JVP normal.

Echocardiogram result discussed; she had not been told of any heart problem before and asks whether it affects her treatment. She also asks whether she should keep taking anastrozole.

Patient goals: wants treatment aimed at cure and hopes to return to full-time work at the school library. Worried about her heart because her mother had heart failure. Husband drives her to all visits.

Case to be presented at lymphoma tumor board 2026-09-18 for a treatment plan; no treatment started yet.

C. Reference answer (hidden from the model)

diagnosis

Diffuse large B-cell lymphoma, NOS, non-germinal-center (non-GCB) phenotype by the Hans algorithm (CD10-, BCL6+, MUM1+). Hans is an immunohistochemical surrogate and is not equivalent to molecular ABC cell-of-origin by gene-expression profiling. EBER negative; CD5 negative. The left axillary node is lymphoma, not recurrent breast carcinoma: cytokeratin AE1/AE3 and GATA3 are negative and there is no epithelial component.

stage

Lugano (Ann Arbor) stage III with B symptoms (documented fevers to 38.4 C and drenching night sweats; weight loss ~3%, below 10%): FDG-avid nodes on both sides of the diaphragm (left axillary and mediastinal above; mesenteric and para-aortic below) with splenic involvement. The spleen is lymphatic tissue (a nodal site, not extranodal), so splenic nodules do not make this stage IV; there is no kidney, adrenal, liver, lung, bone, breast or CNS involvement and the bone marrow biopsy is negative. Largest mass 7.5 cm (mesenteric): below the 10 cm Lugano threshold for bulk, although some trials define bulk at 7.5 cm, so the definition varies.

biology

Double-expressor lymphoma by IHC (MYC 45%, at or above the 40% threshold; BCL2 70%, above the 50% threshold), which carries a somewhat worse prognosis with standard immunochemotherapy. Not double-hit: FISH shows no MYC, BCL2 or BCL6 rearrangement, so this is not high-grade B-cell lymphoma. Double-expressor status alone does not justify DA-EPOCH-R or other intensified regimens. Ki-67 ~80%.

prognosis

IPI 2 of 5 (low-intermediate risk): LDH elevated (510 U/L, ULN 225) and stage III. Age 59 (not >60), ECOG 1 and 0 extranodal sites (the spleen is not counted as an extranodal site) score 0. CNS-IPI 2 of 6 (intermediate risk): LDH and stage only; no kidney/adrenal involvement. Beta-2 microglobulin mildly elevated at 2.9 mg/L.

first_line_treatment

Curative intent. The key issue is cardiac: she received cumulative doxorubicin 240 mg/m2 (dose-dense AC-T, 2016) plus left breast radiotherapy, and LVEF has fallen from 62% (2016) to 45% with GLS -14% and mild global hypokinesis, a >10-point decline to <50% consistent with cancer therapy-related cardiac dysfunction (asymptomatic, NYHA I). Standard full-dose doxorubicin (R-CHOP or Pola-R-CHP) should not be given in this setting. Urgent cardio-oncology referral before cycle 1, with guideline-directed cardioprotection: start an ACE inhibitor or ARB (e.g. switch amlodipine to lisinopril or candesartan) plus a beta-blocker (carvedilol), titrated as blood pressure allows. Recommend curative-intent R-CEOP (rituximab, cyclophosphamide, etoposide, vincristine, prednisone; etoposide substituted for doxorubicin) for 6 cycles as an anthracycline-free regimen; R-GCVP (gemcitabine in place of doxorubicin) is an alternative. Tell the patient that anthracycline-free regimens likely have lower efficacy (cure rates) than anthracycline-based therapy, based on retrospective data. An anthracycline-modified option (R-CHOP with liposomal doxorubicin or with dexrazoxane) is acceptable only if cardio-oncology judges the cardiac risk acceptable after evaluation and optimization. Monitor during therapy with serial echocardiography (LVEF and GLS) every 2-3 cycles and at end of treatment, plus high-sensitivity troponin and NT-proBNP at baseline and with each cycle. DA-EPOCH-R is not indicated: double-expressor status alone does not justify intensification, and it also contains doxorubicin.

cns_risk

Intermediate CNS-relapse risk (CNS-IPI 2). No high-risk anatomic sites: no kidney, adrenal, testicular or breast involvement (the axillary node is lymphoma and the breasts show no uptake), and she has no neurological symptoms. No routine CNS prophylaxis is recommended (no high-dose methotrexate, no intrathecal therapy). MRI brain and lumbar puncture only if neurological symptoms develop.

response_assessment

End-of-treatment FDG PET-CT 6-8 weeks after the last cycle, read with Lugano 2014 criteria and the Deauville 5-point scale (Deauville 1-3 = complete metabolic response). Interim PET-CT after 2-4 cycles is optional and should not change therapy unless it shows progression. Biopsy residual Deauville 4-5 uptake if it would change management. Repeat echocardiogram at end of treatment and during survivorship.

pretreatment_workup

Hepatitis B serology done: HBsAg, anti-HBc and anti-HBs nonreactive, so no HBV reactivation risk from rituximab (consider HBV vaccination). HCV and HIV nonreactive. Echo 2026-09-09: LVEF 45%, GLS -14% (2016: LVEF 62%). Before treatment: cardio-oncology evaluation including ECG, high-sensitivity troponin, NT-proBNP and an ischemic evaluation (coronary CT angiography or stress imaging; left chest radiotherapy raises coronary risk) to define the cause of the LVEF decline; consider cardiac MRI. Bone marrow negative. Creatinine and liver tests normal.

supportive_care

Tumor lysis risk is low-intermediate (LDH ~2.3x ULN, largest mass 7.5 cm, normal creatinine and uric acid): oral hydration and allopurinol starting before cycle 1, with TLS labs at 24-72 hours. Primary G-CSF (pegfilgrastim) with each cycle is reasonable given the febrile neutropenia risk of R-CEOP/R-GCVP and to avoid dose delays. Anastrozole has no meaningful interaction with lymphoma therapy and can continue to complete year 10, or be held during chemotherapy if breast oncology prefers; coordinate with her breast oncologist. Blood pressure (148/90) is controlled with the ACE inhibitor/ARB plus beta-blocker in place of amlodipine. Antiemetics per regimen. Bone health: osteopenia on an aromatase inhibitor plus steroid pulses, continue calcium/vitamin D.

fitness_and_goals

Fit for curative therapy: ECOG 1 (disease-related fatigue and fevers), age 59, normal renal and hepatic function, no heart-failure symptoms (NYHA I). The reduced LVEF changes the regimen, not the curative goal. Her goal is cure and a return to full-time work; explain the trade-off between cardiac safety and the possibly lower efficacy of an anthracycline-free regimen, and decide together with cardio-oncology input. Her worry about her heart (mother with heart failure) is addressed by the cardio-oncology plan and cardioprotection.

missing_information

• Cardio-oncology assessment of the LVEF decline, including its cause: ischemic evaluation (coronary CT angiography or stress imaging), ECG, consider cardiac MRI
• Baseline high-sensitivity troponin and NT-proBNP (not yet drawn)
• Breast oncology input on continuing anastrozole during chemotherapy
• Patient's informed preference after discussing an anthracycline-free regimen versus a cardio-protected anthracycline option

D. Grading rubric (hidden from the model)

Rubric dlbcl_0002.v2.0. Must and should items total 100 points; pass threshold 0.75. A **dangerous** item fails the case (reward 0) when a must item is missed or a must-not item is committed. Points for must-not items are deductions.

ID	Kind	Severity	Pts	Criterion (what the judge reads)
dx_dlbcl_non_gcb	must	important	6	The response states the diagnosis is diffuse large B-cell lymphoma (NOS) with a non-germinal-center (non-GCB) phenotype by the Hans algorithm.
dx_hans_not_abc	should	minor	2	The response notes that Hans non-GCB is an immunohistochemical surrogate and is not equivalent to molecular ABC cell-of-origin by gene-expression profiling.
dx_not_breast_recurrence	should	minor	2	The response states the left axillary node is lymphoma and not recurrent breast carcinoma (cytokeratin AE1/AE3 and GATA3 negative).
stage_iii	must	dangerous	8	The response stages the disease as Lugano/Ann Arbor stage III (nodes on both sides of the diaphragm with splenic involvement).
stage_spleen_lymphatic	should	important	4	The response explains that the spleen is lymphatic tissue (a nodal site), so splenic nodules do not make the disease stage IV, and that there is no non-lymphatic extranodal or marrow involvement.
stage_b_symptoms	should	minor	2	The response notes B symptoms (documented fevers >38 C and drenching night sweats).
stage_bulk_definition	should	minor	2	The response notes the 7.5 cm mass is below the 10 cm Lugano bulk threshold, acknowledging that bulk definitions vary (some trials use 7.5 cm).
mn_stage_iv	must_not	dangerous	0	The response stages this patient as stage IV (e.g. by counting splenic involvement as extranodal disease).
mol_del_not_dh	must	important	6	The response classifies the lymphoma as double-expressor by IHC (MYC 45% >=40%, BCL2 70% >=50%) but NOT double-hit (FISH shows no MYC, BCL2 or BCL6 rearrangement).
mn_calls_double_hit	must_not	dangerous	0	The response classifies this lymphoma as double-hit / high-grade B-cell lymphoma with MYC and BCL2 rearrangements.
mn_da_epoch	must_not	important	5	The response recommends DA-EPOCH-R (or another intensified regimen) on the basis of double-expressor biology.
prog_ipi_2	must	important	6	The response calculates IPI as 2 (low-intermediate risk): elevated LDH and stage III; age 59, ECOG 1 and no extranodal sites (spleen is not counted as extranodal) score zero.
prog_cns_ipi_2	should	important	3	The response calculates CNS-IPI as 2 (intermediate risk).
cns_no_routine_prophylaxis	should	important	4	The response states that routine CNS prophylaxis is not indicated (CNS-IPI 2, no kidney/adrenal, testicular or breast involvement).
mn_mandates_cns_prophylaxis	must_not	important	4	The response mandates CNS prophylaxis (high-dose methotrexate or intrathecal therapy) for this patient.
tx_cardiac_aware_no_full_dox	must	dangerous	10	The response recognizes anthracycline-related cardiac risk (prior doxorubicin 240 mg/m2, left chest radiotherapy, LVEF fall from 62% to 45%, i.e. cancer therapy-related cardiac dysfunction) and does NOT plan standard full-dose doxorubicin (R-CHOP or Pola-R-CHP) without first addressing cardiac function.
mn_standard_dox_ignores_heart	must_not	dangerous	0	The response recommends standard R-CHOP or Pola-R-CHP (full-dose doxorubicin) without conditioning it on the reduced LVEF / prior anthracycline exposure (e.g. without cardio-oncology clearance, a cardioprotective modification, or a non-anthracycline alternative).
tx_cardio_onc_referral	must	important	6	The response requests urgent cardio-oncology (or cardiology) evaluation before starting therapy.
tx_cardioprotection	must	important	5	The response starts or optimizes guideline-directed cardioprotective therapy: an ACE inhibitor/ARB (or ARNI) plus a beta-blocker (e.g. carvedilol), for example replacing amlodipine.

ID	Kind	Severity	Pts	Criterion (what the judge reads)
tx_anthracycline_free_curative	must	important	8	The response recommends a curative-intent anthracycline-free or anthracycline-modified regimen, e.g. R-CEOP (etoposide in place of doxorubicin) or R-GCVP; R-CHOP with liposomal doxorubicin or dexrazoxane is acceptable only if conditioned on cardio-oncology judging it safe.
tx_lower_efficacy_ack	should	minor	2	The response acknowledges that anthracycline-free regimens likely have lower efficacy than anthracycline-based therapy.
tx_cardiac_monitoring	should	important	4	The response plans serial cardiac monitoring during therapy: repeat LVEF/GLS imaging plus troponin and/or natriuretic peptide (BNP/NT-proBNP).
mn_non_curative	must_not	dangerous	0	The response recommends palliative-only treatment, best supportive care, or hospice as the plan.
resp_eot_pet_lugano	must	important	6	The response plans an end-of-treatment FDG PET-CT read with Lugano criteria / Deauville 5-point scale.
sup_tls_allopurinol	should	minor	2	The response assesses tumor lysis risk as low-intermediate (normal creatinine and uric acid, LDH ~2x ULN) and gives allopurinol with hydration for cycle 1.
sup_gcsf	should	minor	2	The response recommends (or reasonably considers) primary G-CSF support with each cycle.
sup_anastrozole	should	minor	1	The response addresses the ongoing anastrozole: continuing it is acceptable (no meaningful interaction), or holding it in coordination with breast oncology is also acceptable.
work_hbv_reviewed	should	minor	2	The response confirms hepatitis B serology was checked (HBsAg and anti-HBc nonreactive) before rituximab.
fit_curative_ecog1	must	important	4	The response judges the patient fit for curative-intent therapy (ECOG 1, normal organ function apart from the LVEF) and acknowledges her goal of cure.
miss_cardiac_workup	should	important	3	The response lists the cause of the LVEF decline / cardio-oncology work-up as missing (e.g. ischemic evaluation given prior left chest radiotherapy, cardiac MRI, baseline troponin and NT-proBNP).

E. Grader self-tests (regex mode, recomputed now)

Each adversarial answer must produce its expected verdict. 01_reference is the gold answer.

Answer	Reward	Passed	Expected	Dangerous errors fired
01_reference	1.00	yes	pass	-
bad_double_hit	0.00	no	fail	mn_calls_double_hit
bad_pola_despite_lvef_elsewhere	0.00	no	fail	mn_standard_dox_ignores_heart
bad_stage_iv	0.00	no	fail	stage_iii, mn_stage_iv
bad_standard_rchop_ignores_lvef	0.00	no	fail	tx_cardiac_aware_no_full_dox, mn_standard_dox_ignores_heart
good_conditional_rchop_cardiac_aware	1.00	yes	pass	-
v1_gold_answer	0.00	no	fail	tx_cardiac_aware_no_full_dox, mn_standard_dox_ignores_heart

F. Reviewer sign-off

Reviewer	Credentials	Date	Verdict	Changes requested